

Question number	Answer	Additional guidance	Mark
9(a)	- mass of carbon in gt calculated in one year (1) 1.13×10^{13} (kg) (1) OR - mass in gt converted into mass in kg (1) 1.13×10^{13} (kg) (1)	11.3 (gt per year) $113 \times 10^{12} / 1.13 \times 10^{14} / 11.3 \times 10^{13}$	(2)

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9(b)	A description that includes four of the following points: - GALP converted into glucose (1) - glucose and fructose used to produce sucrose (in the leaves) (1) {sucrose / amino acids} moved (to the roots) in the phloem (1) {glucose / sucrose / amino acids} used to make {organic molecules / named organic molecule / named cell component} (in the roots) (1) - credit details of how a named polymer is synthesised (1)	DO NOT ACCEPT glucose / protein DO NOT ACCEPT chlorophyll e.g. calcium ions used to make (calcium pectate / cell walls), condensation reaction to form starch	(4)

Question number					Mark																								
9(c)	<table><tr><th rowspan="2">Molecule</th><th colspan="4">Inorganic ion</th></tr><tr><th>nitrate only</th><th>phosphate only</th><th>both nitrate and phosphate</th><th>neither nitrate nor phosphate</th></tr><tr><td>cellulose</td><td>[x]</td><td>[x]</td><td>[x]</td><td>[x]</td></tr><tr><td>nucleic acid</td><td>[x]</td><td>[x]</td><td>[x]</td><td>[x]</td></tr><tr><td>triglyceride</td><td>[x]</td><td>[x]</td><td>[x]</td><td>[x]</td></tr></table>	Molecule	Inorganic ion				nitrate only	phosphate only	both nitrate and phosphate	neither nitrate nor phosphate	cellulose	[x]	[x]	[x]	[x]	nucleic acid	[x]	[x]	[x]	[x]	triglyceride	[x]	[x]	[x]	[x]				
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(3)																													

Question	Answer	Mark
*9(d)	Yearly mean temperature: • (overall) has the greatest influence /has the greatest influence in forests and shrublands • because temperature affects enzyme activity • details of effects • in plants of all three ecosystems • therefore faster photosynthesis and production of glucose • RUBISCO works faster increasing light-independent reactions • and production of GALP Soil moisture content: • has the greatest influence in grasslands / second greatest influence (overall) • because water needed for photolysis / light-dependent reactions • without which photosynthesis will not occur and GALP will not be made • water needed for transpiration and transport of mineral ions • water needed to transport sucrose / amino acids to roots • water maybe more important to trees / grass as they have extensive root systems Density of trees: • influential in all ecosystems • high density of trees will cause {competition for light / shading} • so rate of light-dependent reactions will be slower • so less ATP for light-independent reactions • so less biomass available in the roots Carbon : nitrogen ratio: • affects all three types of plant to some extent but low importance • plants need soil nitrates for amino acid production • less nitrogen less protein available • protein examples e.g. enzymes, RUBISCO • for active transport of mineral ions Depth of soil: • affects trees the most but least importance • soil needed for anchorage / water / mineral ions • because they need deeper soil and more extensive root system • therefore more biomass needed in their roots • shrubs / grass can grow in shallow soil as root system less extensive	(6)

			Additional guidance
vel 0	0	No awardable content	
vel 1	1-2	Demonstrates isolated elements of biological knowledge and understanding to the given context with generalised comments made. Vague statements related to consequences are made with limited linkage to a range of scientific ideas, processes, techniques and procedures. The discussion will contain basic information with some attempt made to link knowledge and understanding to the given context.	Simple links made between factors and the ecosystems 1 mark = simple comment about {one factor / one ecosystem} 2 marks = simple comments about {three factors / three ecosystems}
vel 2	3-4	Demonstrates adequate knowledge and understanding by selecting and applying some relevant biological facts / concepts. Consequences are discussed which are occasionally supported through linkage to a range of scientific ideas, processes, techniques and procedures. The discussion shows some linkages and lines of scientific reasoning with some structure.	Extended links made between factors and ecosystems 3 marks = extended comments on one factor 4 marks = extended comments on two factors
vel 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting and applying relevant biological facts / concepts. Consequences are discussed which supported throughout by sustained linkage to a range of scientific ideas, processes, techniques and procedures. The discussion shows a well-developed and sustained line of scientific reasoning which is clear and logically structured.	Extended links made between factors and ecosystems that includes discussion about the relative importance of factors in root biomass in the different ecosystems 5 marks = extended comments on three factors 6 marks = extended comments on four factors

